

## FORMULÁRIO

### Derivadas

|                                                                  |                                                                   |                                                                    |
|------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------|
| $(uv)' = u'v + uv'$                                              | $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$               | $(u^\alpha)' = \alpha u^{\alpha-1}u', \quad \alpha \in \mathbb{R}$ |
| $(a^u)' = u'a^u \ln a, \quad a \in \mathbb{R}^+ \setminus \{1\}$ | $(e^u)' = u'e^u$                                                  | $(u^v)' = vu^{v-1}u' + u^v v' \ln u$                               |
| $(\log_a u)' = \frac{u'}{u \ln a}$                               | $(\ln u)' = \frac{u'}{u}$                                         | $(\sin u)' = u' \cos u$                                            |
| $(\cos u)' = -u' \sin u$                                         | $(\tan u)' = u' \sec^2 u$                                         | $(\cotan u)' = -u' \operatorname{cosec}^2 u$                       |
| $(\sec u)' = u' \sec u \tan u$                                   | $(\operatorname{cosec} u)' = -u' \operatorname{cosec} u \cotan u$ |                                                                    |
| $(\arcsin u)' = \frac{u'}{\sqrt{1-u^2}}$                         | $(\arccos u)' = \frac{-u'}{\sqrt{1-u^2}}$                         | $(\arctan u)' = \frac{u'}{1+u^2}$                                  |

### Primitivas Imediatas

|                                                                                 |                                                                                   |                                            |
|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------|
| $\int u' u^\alpha dx = \frac{u^{\alpha+1}}{\alpha+1} + C, \quad \alpha \neq -1$ | $\int u' a^u dx = \frac{a^u}{\ln a} + C$                                          | $\int u' e^u dx = e^u + C$                 |
| $\int \frac{u'}{u} dx = \ln  u  + C$                                            | $\int u' \sin u dx = -\cos u + C$                                                 | $\int u' \cos u dx = \sin u + C$           |
| $\int u' \tan u dx = -\ln  \cos u  + C$                                         | $\int u' \cotan u dx = \ln  \sin u  + C$                                          |                                            |
| $\int u' \sec^2 u dx = \tan u + C$                                              | $\int u' \operatorname{cosec}^2 u dx = -\cotan u + C$                             |                                            |
| $\int u' \sec u dx = \ln  \sec u + \tan u  + C$                                 | $\int u' \operatorname{cosec} u dx = \ln  \operatorname{cosec} u - \cotan u  + C$ |                                            |
| $\int u' \sec u \tan u dx = \sec u + C$                                         | $\int u' \operatorname{cosec} u \cotan u dx = -\operatorname{cosec} u + C$        |                                            |
| $\int \frac{u'}{\sqrt{1-u^2}} dx = \arcsin u + C$                               | $\int \frac{-u'}{\sqrt{1-u^2}} dx = \arccos u + C$                                | $\int \frac{u'}{1+u^2} dx = \arctan u + C$ |

### Substituições

|                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------|
| $FR(x, x^{p/q}, x^{r/s}, \dots) \longrightarrow x = t^m, \quad m = \operatorname{mmc}(q, s, \dots)$                                      |
| $FR(x, (ax+b)^{p/q}, (ax+b)^{r/s}, \dots) \longrightarrow ax+b = t^m, \quad m = \operatorname{mmc}(q, s, \dots)$                         |
| $FR(a^{rx}, a^{sx}, \dots) \longrightarrow a^{mx} = t, \quad m = \operatorname{mdc}(r, s, \dots)$                                        |
| $FR\left(x, \sqrt{a^2 - b^2 x^2}\right) \longrightarrow x = \frac{a}{b} \sin t$                                                          |
| $FR\left(x, \sqrt{a^2 + b^2 x^2}\right) \longrightarrow x = \frac{a}{b} \tan t$                                                          |
| $FR(\sin x, \cos x) \longrightarrow t = \tan\left(\frac{x}{2}\right); \sin x = \frac{2t}{1+t^2} \text{ e } \cos x = \frac{1-t^2}{1+t^2}$ |

### Fórmulas Trigonométricas

$$\sin(2x) = 2 \sin x \cos x$$

$$\cos(2x) = \cos^2 x - \sin^2 x$$

$$\sin^2 x = \frac{1 - \cos(2x)}{2}$$

$$\cos^2 x = \frac{1 + \cos(2x)}{2}$$

$$\sec^2 x = 1 + \tan^2 x$$

$$\operatorname{cosec}^2 x = 1 + \cotan^2 x$$